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In [15]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the dataset
# The dataset does not have a header row, so we explicitly define the column names
column_names = ['Tweet_ID', 'Entity', 'Sentiment', 'Tweet_Content']
df = pd.read_csv(r'C:\Users\LOQ\Downloads\twitter_training.csv\twitter_training.csv')

# 2. Data Cleaning
print("--- Initial Data Info ---")
print(df.info())

# Drop rows where the tweet content or sentiment label is missing
df.dropna(subset=['Tweet_Content', 'Sentiment'], inplace=True)

# Clean text: Remove duplicate tweets to avoid skewing the sentiment patterns
df.drop_duplicates(subset=['Tweet_Content'], inplace=True)

print("\n--- Data Info After Cleaning ---")
print(df.info())

# 3. Exploratory Data Analysis (EDA)
# Global Sentiment Distribution Count
sentiment_counts = df['Sentiment'].value_counts()
print("\n--- Global Sentiment Counts ---")
print(sentiment_counts)

# 4. Visualizations

# Chart 1: Global Sentiment Distribution (Bar Plot)
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Sentiment', order=sentiment_counts.index)
plt.title('Overall Sentiment Distribution on Twitter')
plt.xlabel('Sentiment Class')
plt.ylabel('Number of Tweets')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.savefig('global_sentiment_distribution.png')
plt.show()

# Chart 2: Top 10 Most Mentioned Entities/Brands
top_entities = df['Entity'].value_counts().head(10).index
df_top_entities = df[df['Entity'].isin(top_entities)]

plt.figure(figsize=(12, 6))
sns.countplot(data=df_top_entities, y='Entity', order=top_entities)
plt.title('Top 10 Most Mentioned Brands/Entities')
plt.xlabel('Number of Tweets')
plt.ylabel('Entity')
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.savefig('top_entities.png')
plt.show()

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# Chart 3: Sentiment Breakdown Per Top Entity (Segmented Analysis)
plt.figure(figsize=(14, 8))
sns.countplot(data=df_top_entities, y='Entity', hue='Sentiment', order=top_entities)
plt.title('Sentiment Breakdown Across Top 10 Brands/Entities')
plt.xlabel('Number of Tweets')
plt.ylabel('Entity')
plt.legend(title='Sentiment', loc='lower right')
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig('sentiment_by_brand.png')
plt.show()

```

--- Initial Data Info ---

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74682 entries, 0 to 74681
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Tweet_ID        74682 non-null  int64
1   Entity          74682 non-null  object
2   Sentiment       74682 non-null  object
3   Tweet_Content   73996 non-null  object
dtypes: int64(1), object(3)
memory usage: 2.3+ MB
None

```

--- Data Info After Cleaning ---

```

<class 'pandas.core.frame.DataFrame'>
Index: 69491 entries, 0 to 74681
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Tweet_ID        69491 non-null  int64
1   Entity          69491 non-null  object
2   Sentiment       69491 non-null  object
3   Tweet_Content   69491 non-null  object
dtypes: int64(1), object(3)
memory usage: 2.7+ MB
None

```

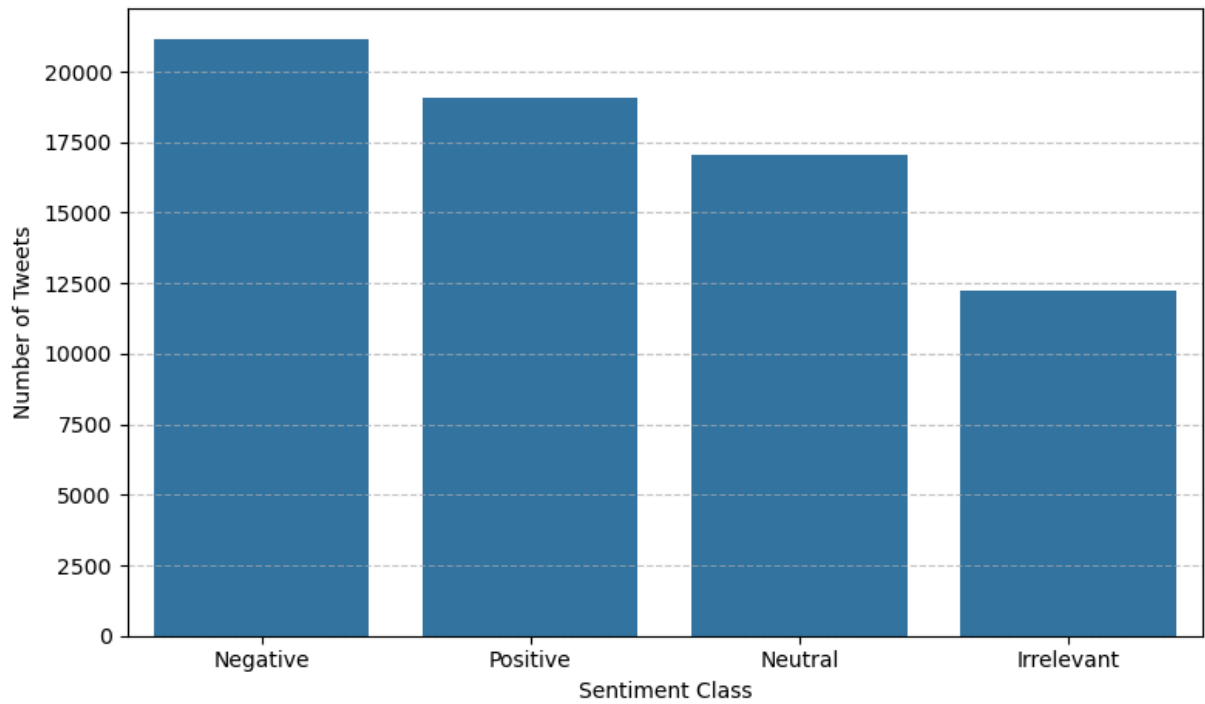
--- Global Sentiment Counts ---

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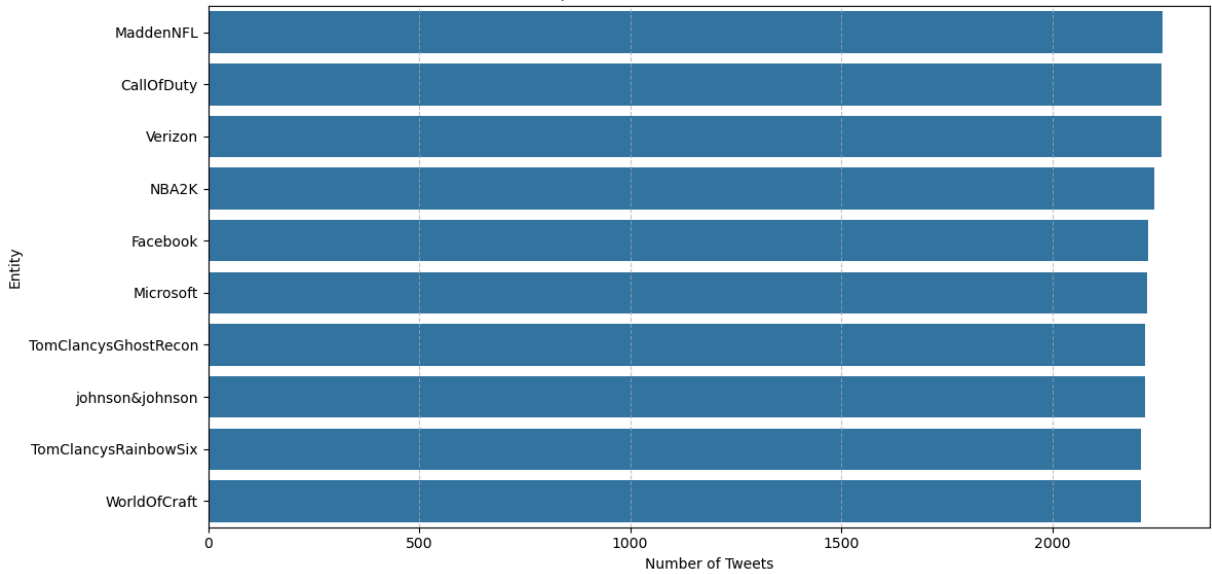
Sentiment
Negative      21166
Positive      19067
Neutral       17042
Irrelevant    12216
Name: count, dtype: int64

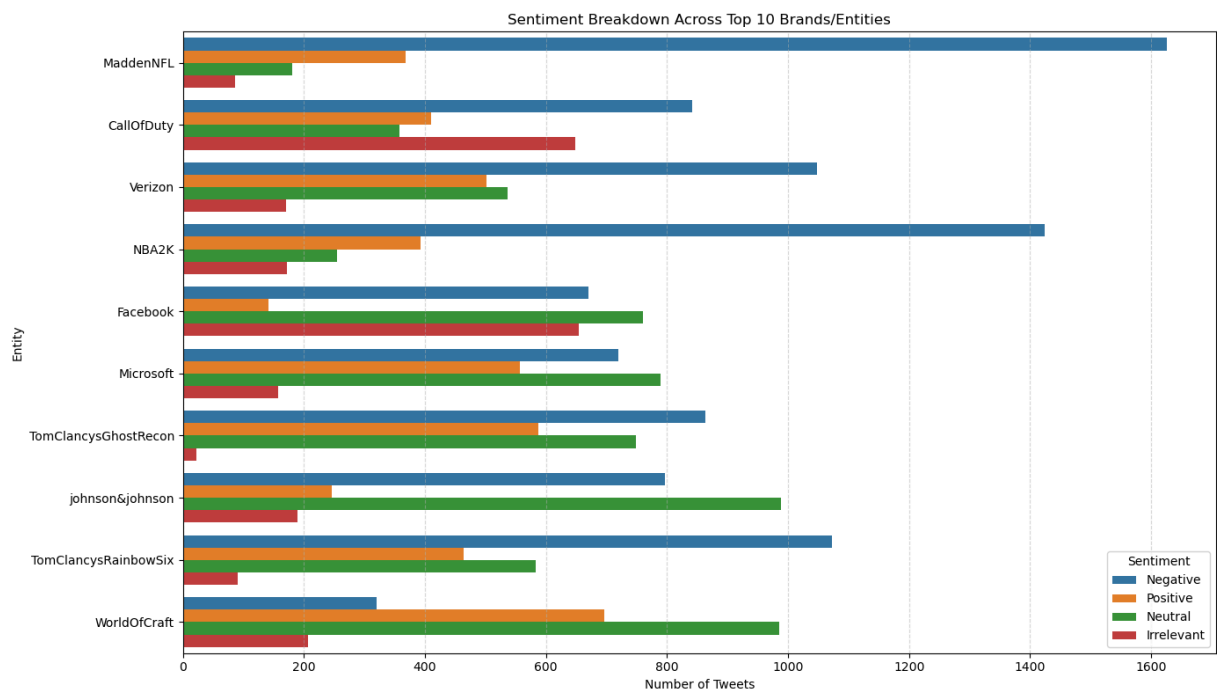
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Overall Sentiment Distribution on Twitter



Top 10 Most Mentioned Brands/Entities





In []: